

UUV MuJoCo v2.2 current 모델 발표

MuJoCo built-in ellipsoid fluid + Python thrust/hydrostatics
latest actual rosbag 기준 current 모델 설명

2. 목적과 범위

- 이번 발표는 current v2.2 모델만 설명
- legacy는 구조 차이만 짧게 언급
- 핵심 질문: built-in과 custom이 어디서 갈리고, thrust와 buoyancy를 왜 이렇게 구현했는가
- 기준 데이터: latest actual rosbag + current repository snapshot

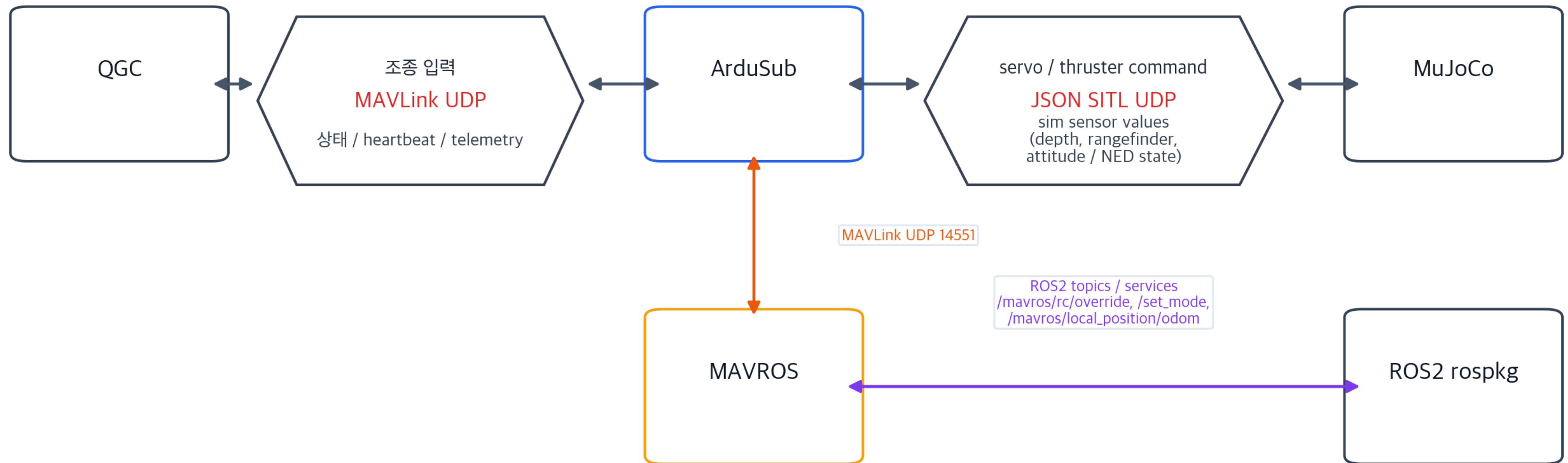
current actual 평균 path 0.122 m / horizontal 0.0273 m

current actual 평균 pitch 0.0885 deg / roll 0.0809 deg

3. 시스템 전체 구조

QGC / ArduSub / MuJoCo / ROS2 데이터 흐름

UUV MuJoCo v2.2 정식 호환 시스템 구조



4. current 물리엔진 구조

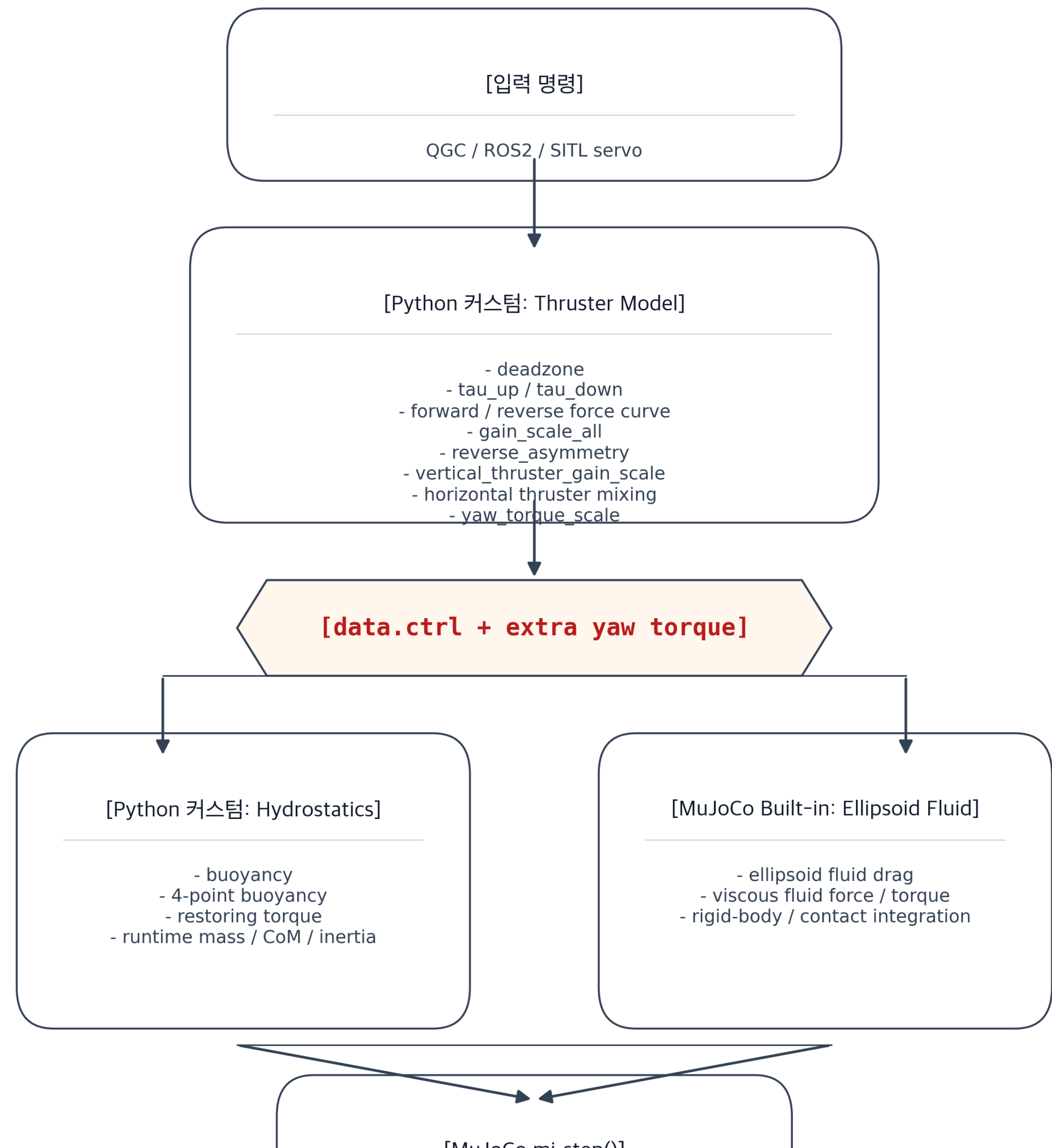
- MuJoCo built-in

- ellipsoid fluid drag
- viscous fluid force / torque
- rigid-body integration, mj_step()

- Python custom

- thrust shaping and mixing
- buoyancy / restoring torque
- runtime mass / CoM / inertia

- 핵심: drag proxy와 buoyancy point를 분리해서 씴



5. current thrust handling path

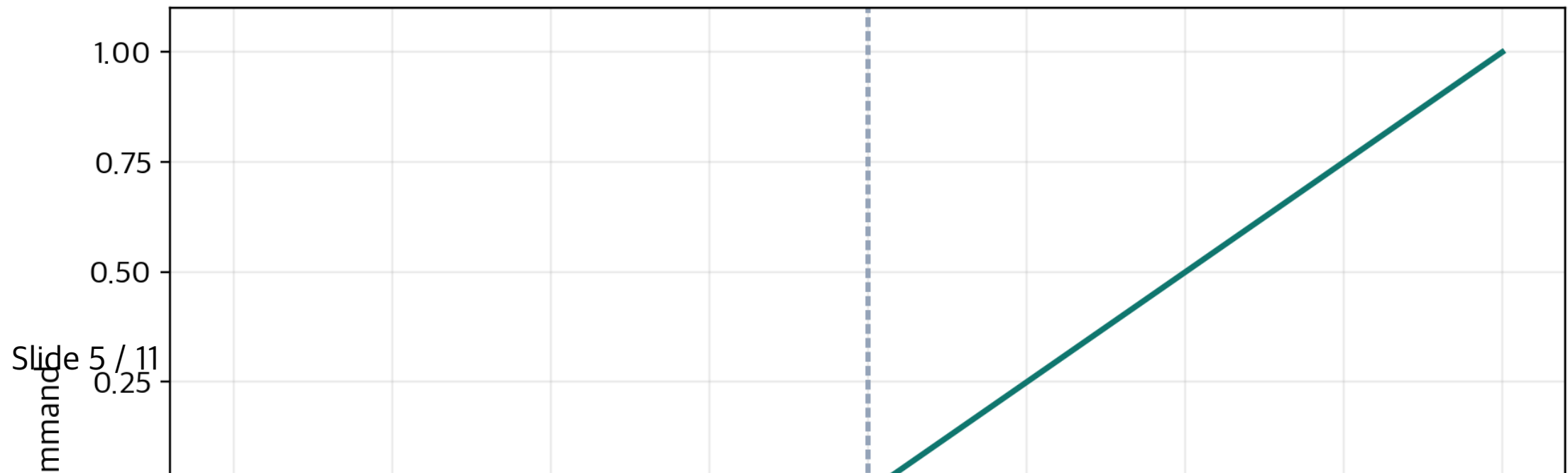
- normalized command -> lag -> deadzone / clamp -> force curve -> data.ctrl
- vertical_thruster_gain_scale, yaw_torque_scale가 뒤에서 추가 보정
- current default 경로는 simple polynomial thrust model

Current thrust handling path

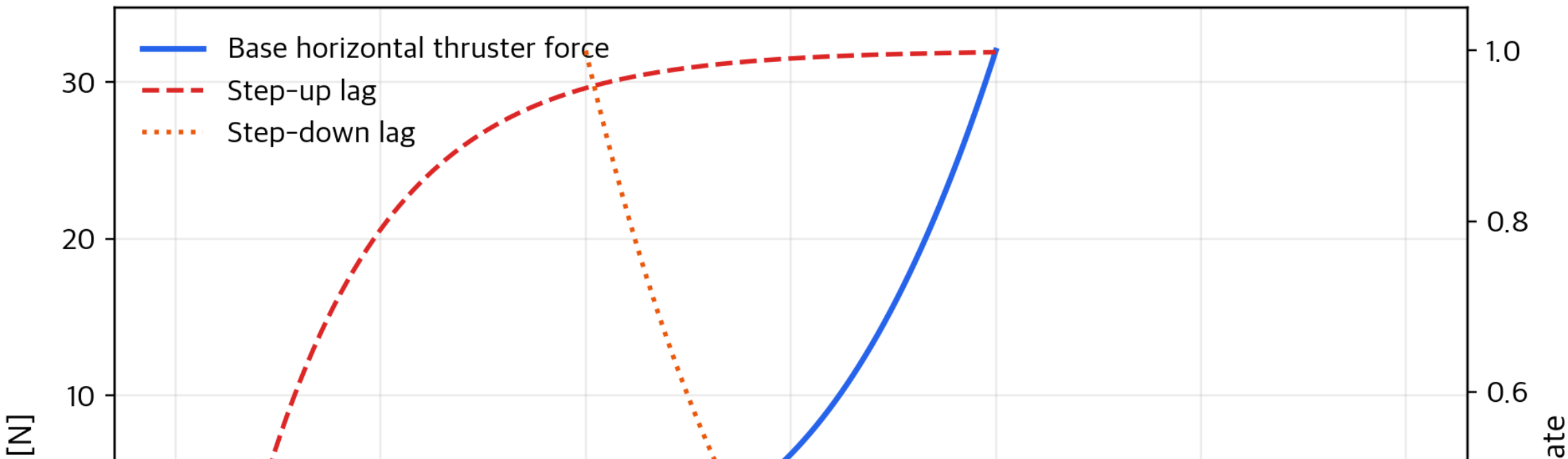


Current default path uses the simple polynomial model.
Performance-table interpolation is not the default SITL path.

Command shaping

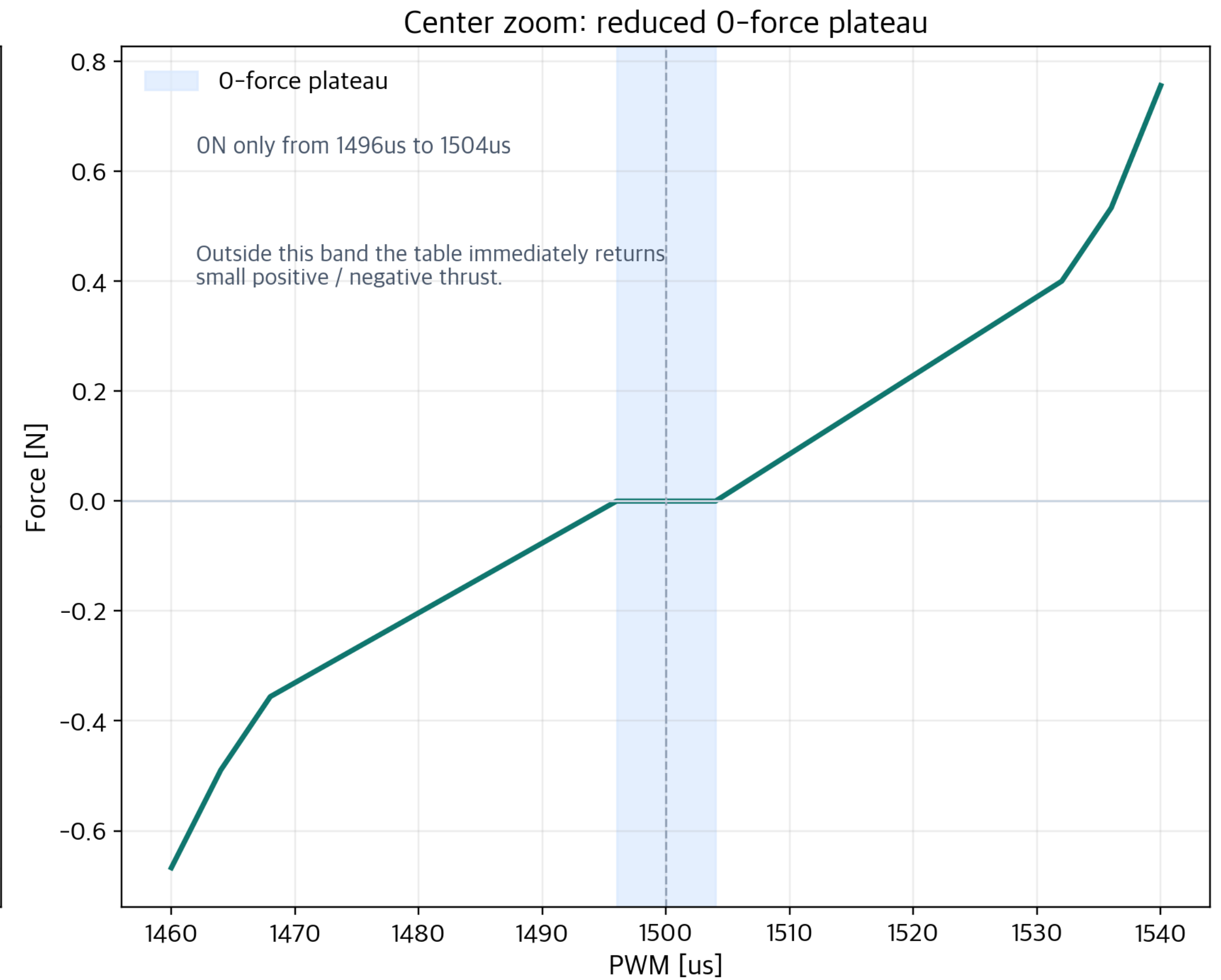
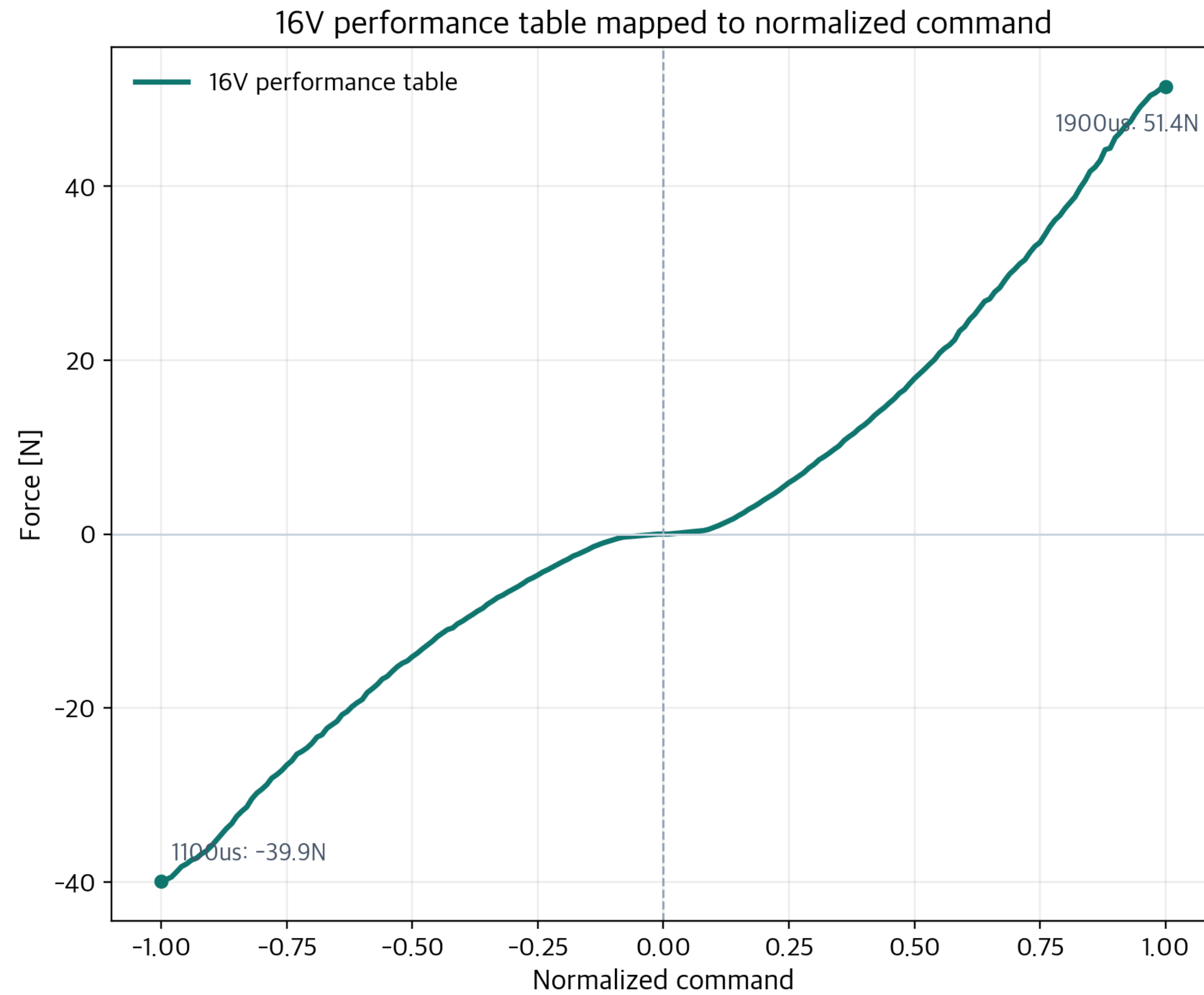


Force curve and lag behavior



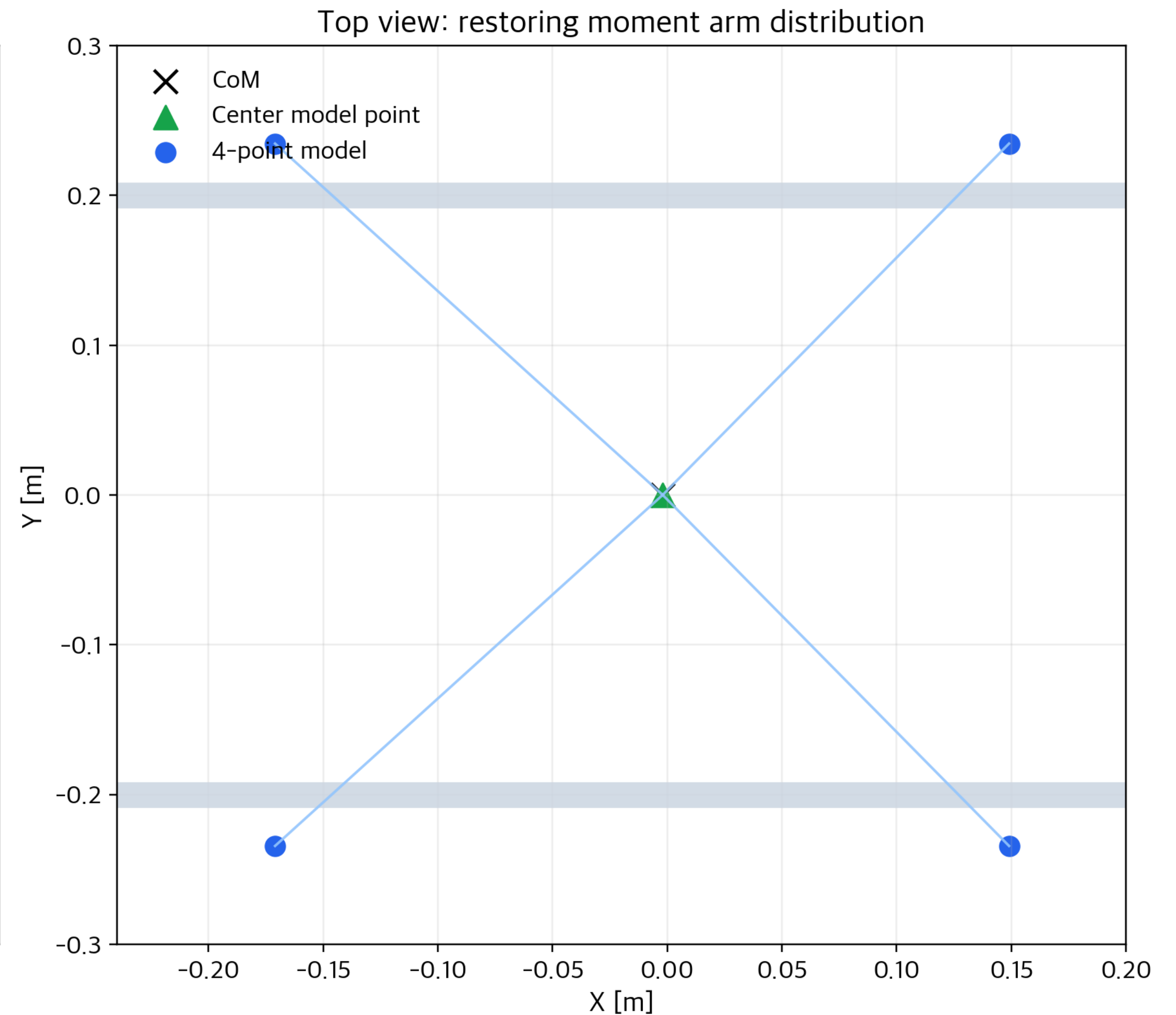
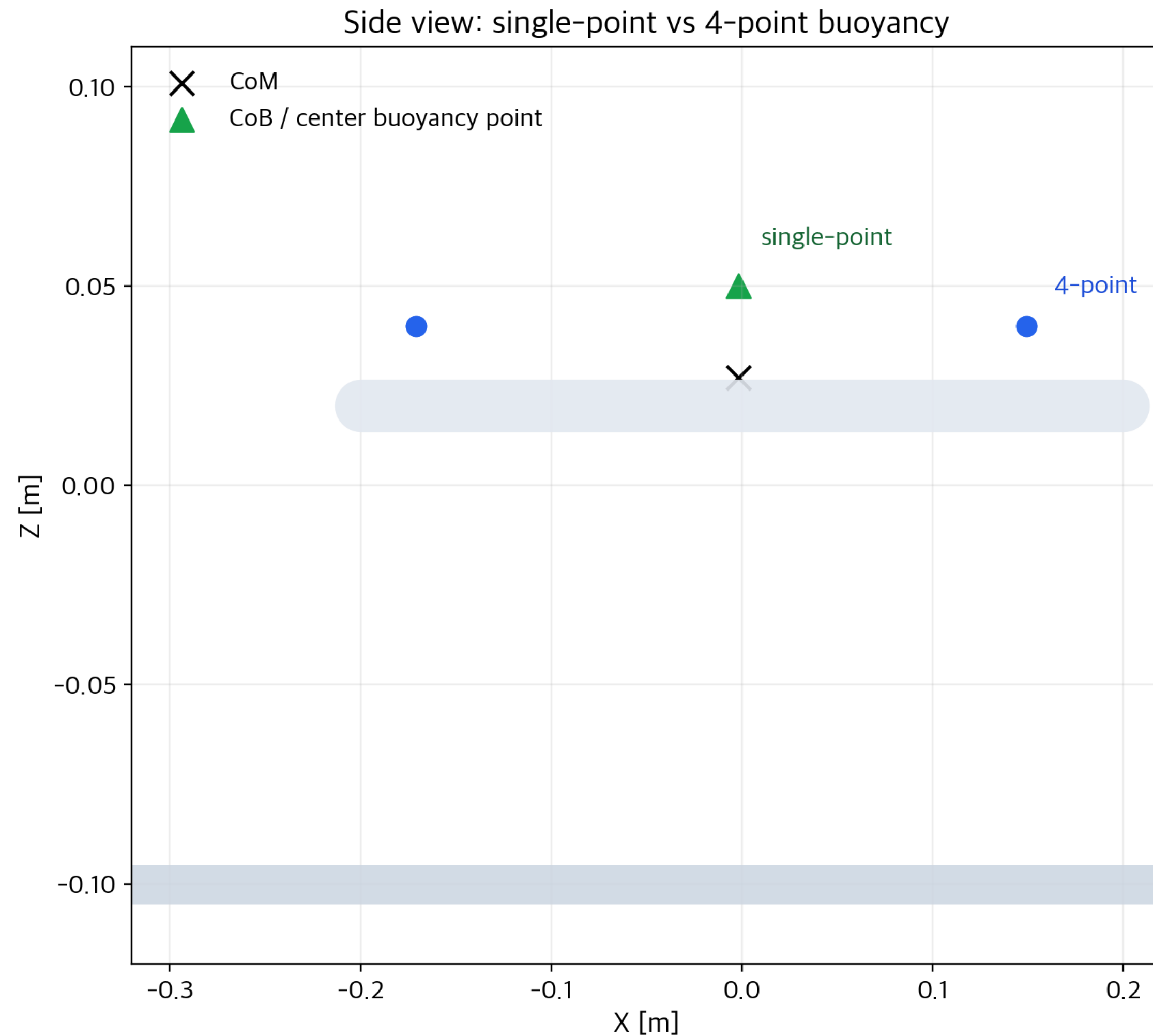
6. Thruster performance reference

- 좌측: 16V performance table 전체 곡선을 normalized command로 표현
- 우측: 1500us 근처 center zoom
- 핵심: 0-force plateau만 줄이고 1900us 끝단 값은 유지



7. buoyancy model design

- single-point center: 총 buoyancy를 한 점에 모아서 적용
- current 4-point: 같은 총 buoyancy를 4개 upper point에 분산
- 기대 효과: 과한 fake pitch 감소, restoring behavior 개선, heave/attitude 응답 안정화

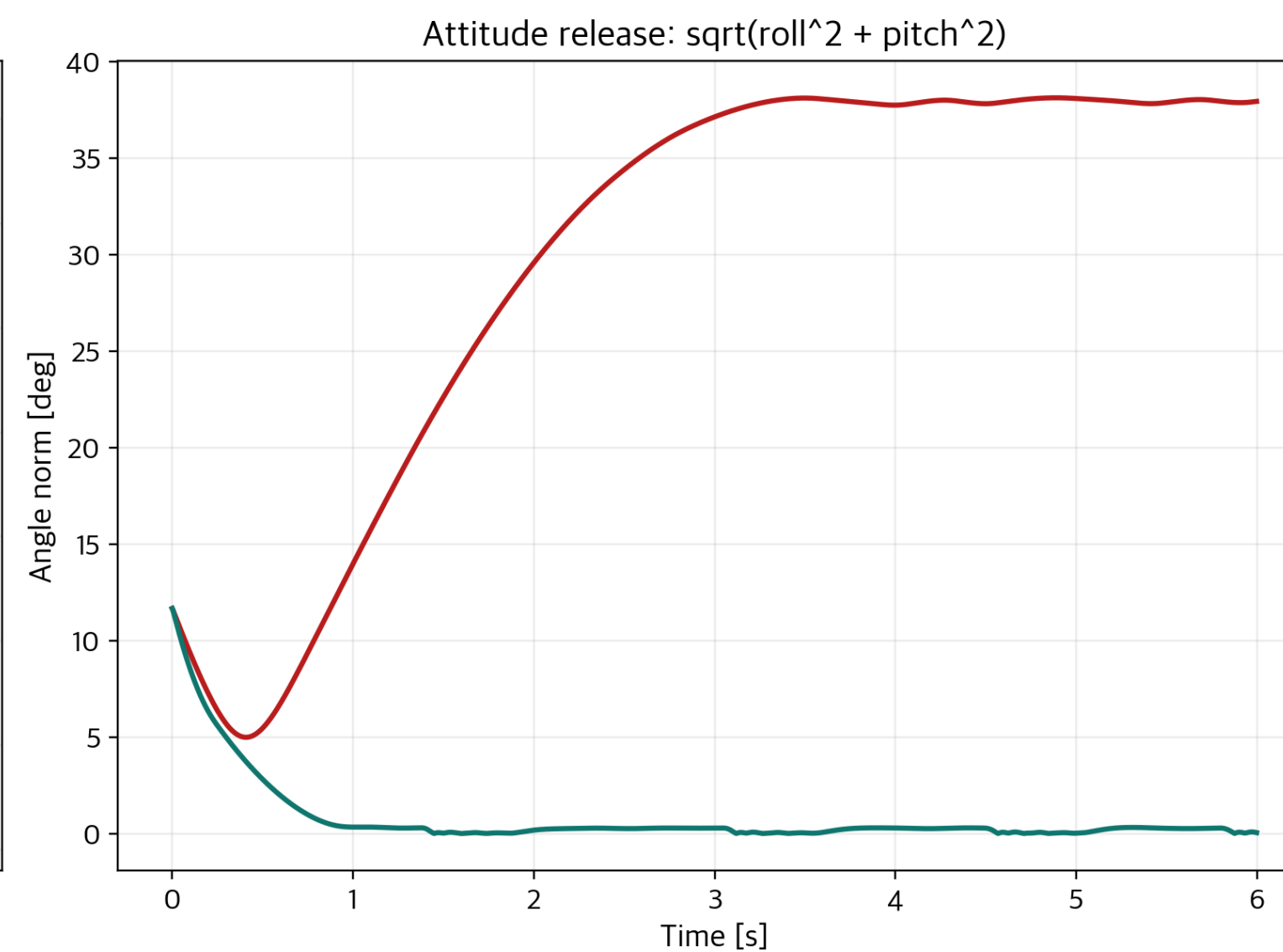
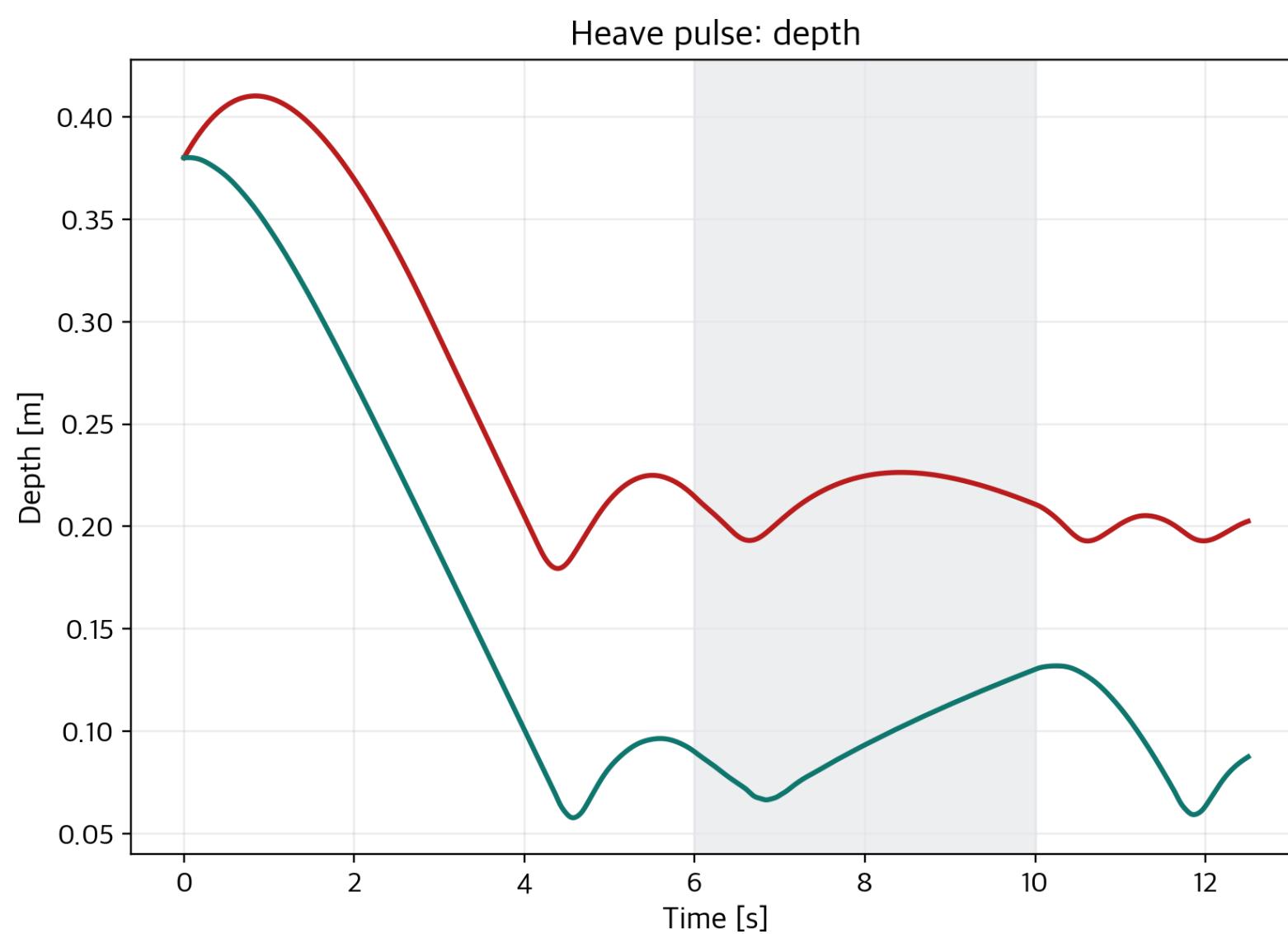
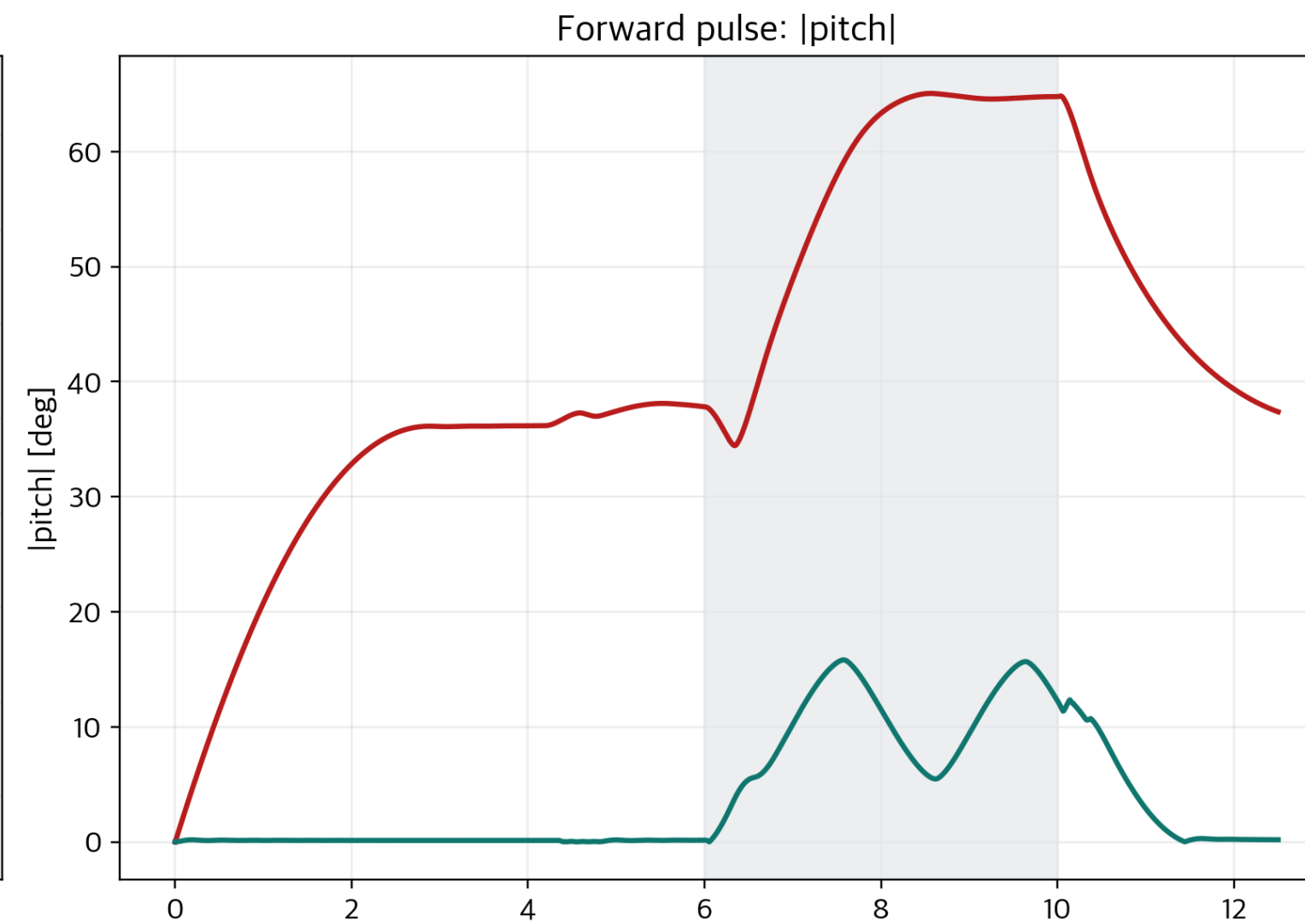
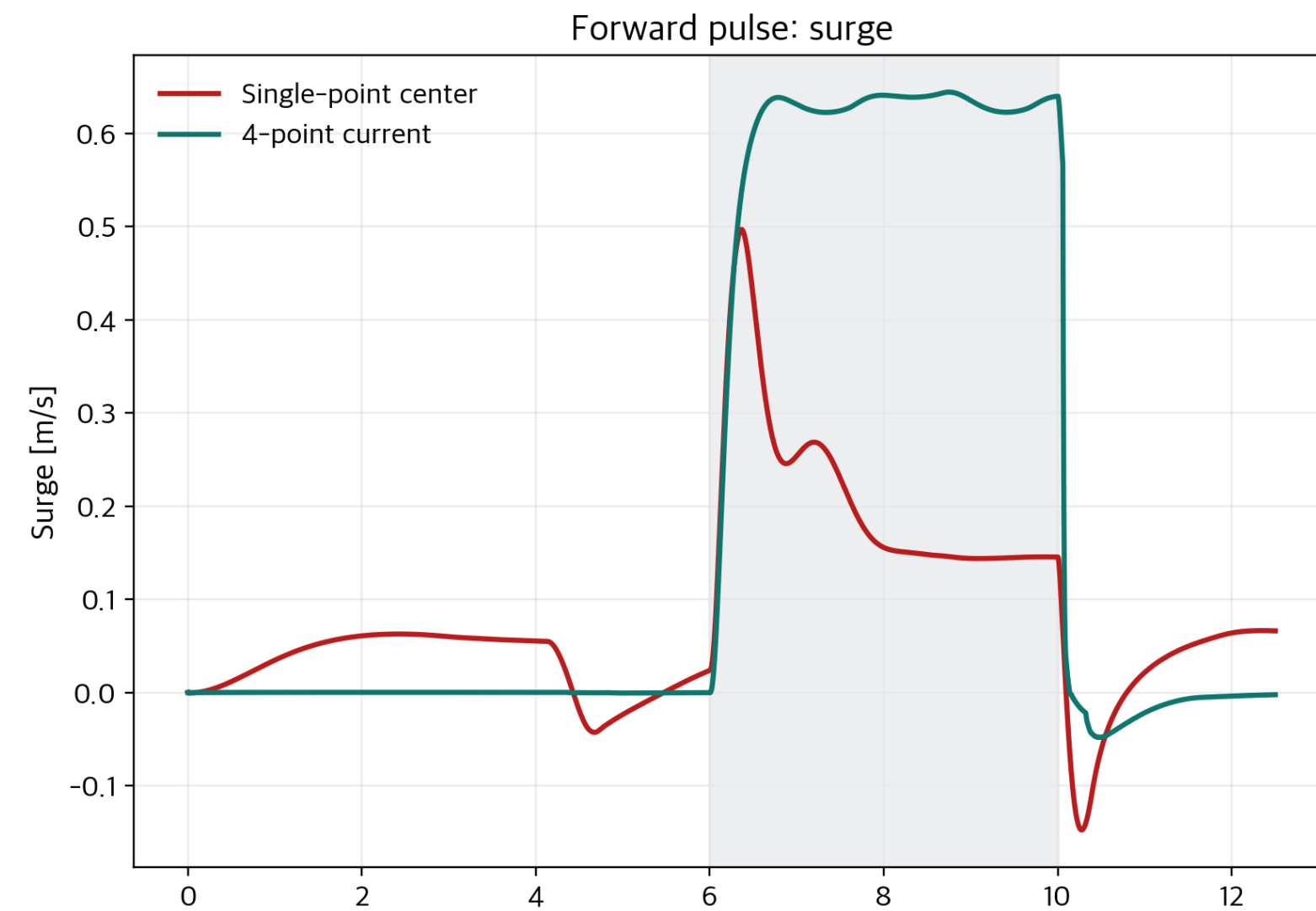


8. center vs 4-point response 비교

- same current scene / same thrust model / same scripted harness

- single-point center는 forward에서 과한 pitch가 남음

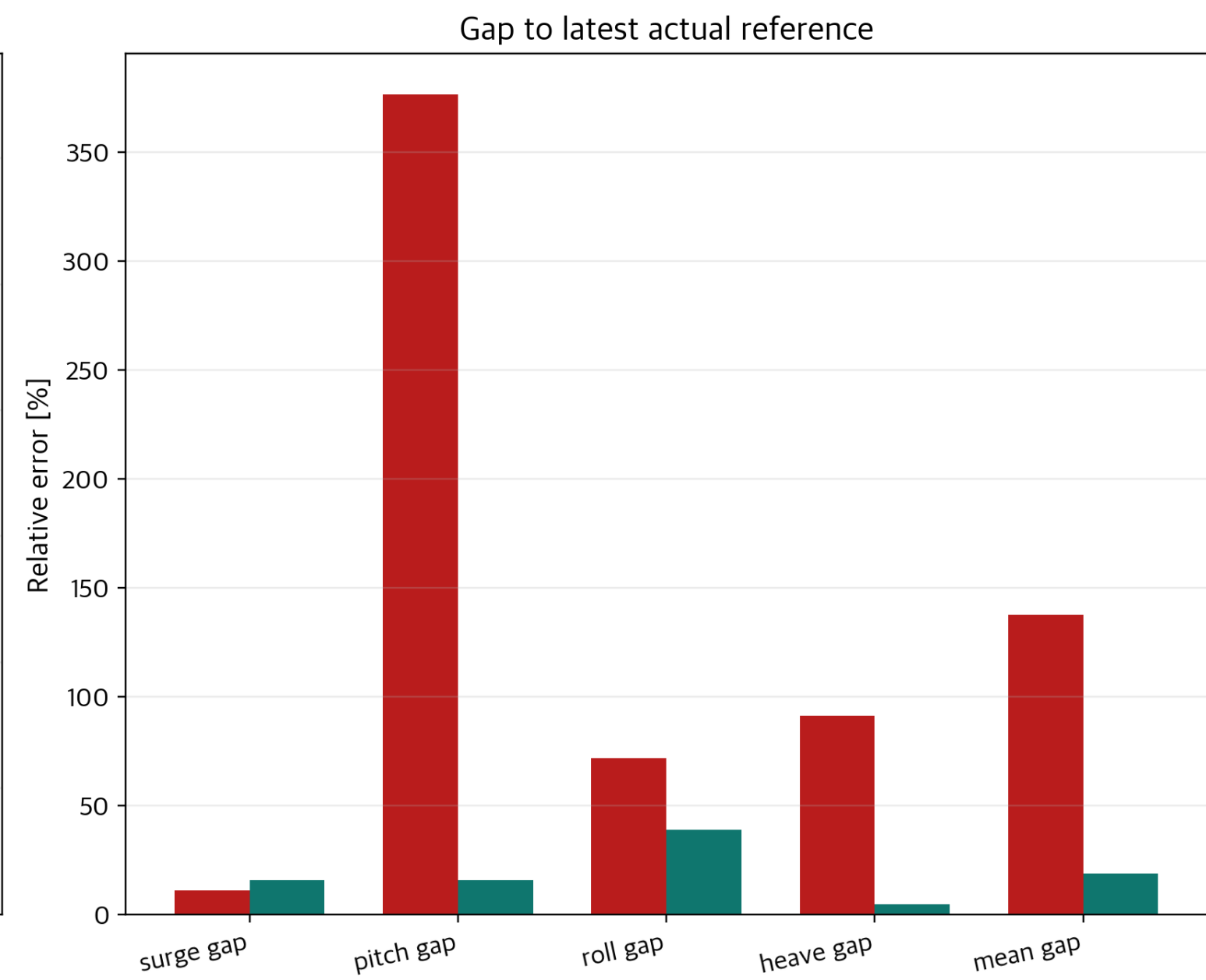
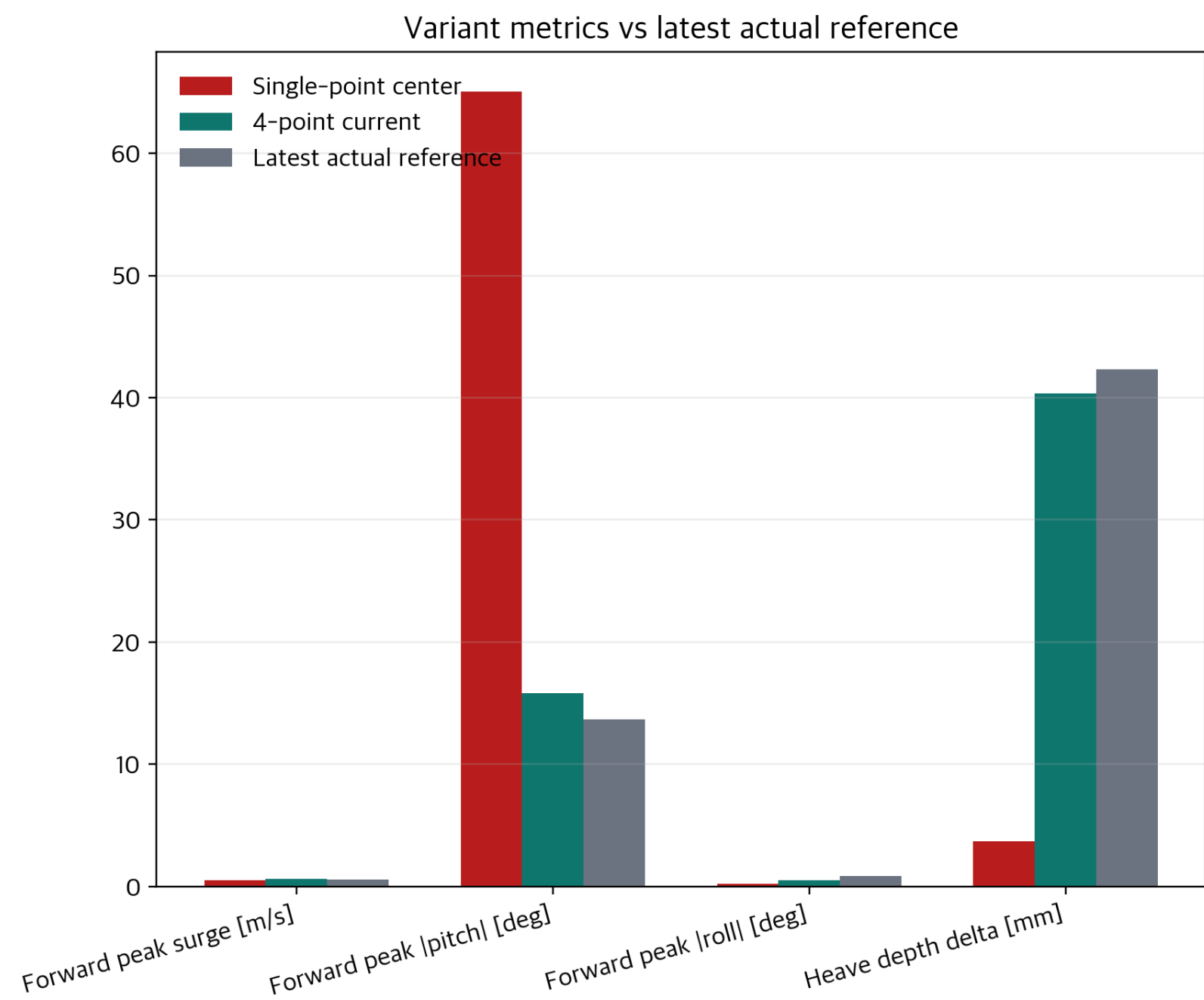
- 4-point는 release 안정성과 forward posture가 훨씬 정돈됨



4-point는 release stability와 forward posture에서 확실히 더 정돈됨

9. similarity metrics

• single-point center gap index:
137.61%



- 4-point forward surge 0.644 m/s
- 4-point forward pitch 15.81 deg
- 4-point heave delta 40.33 mm
- 결론: 4-point buoyancy는 cosmetic이 아니라 similarity를 만드는 핵심 설계

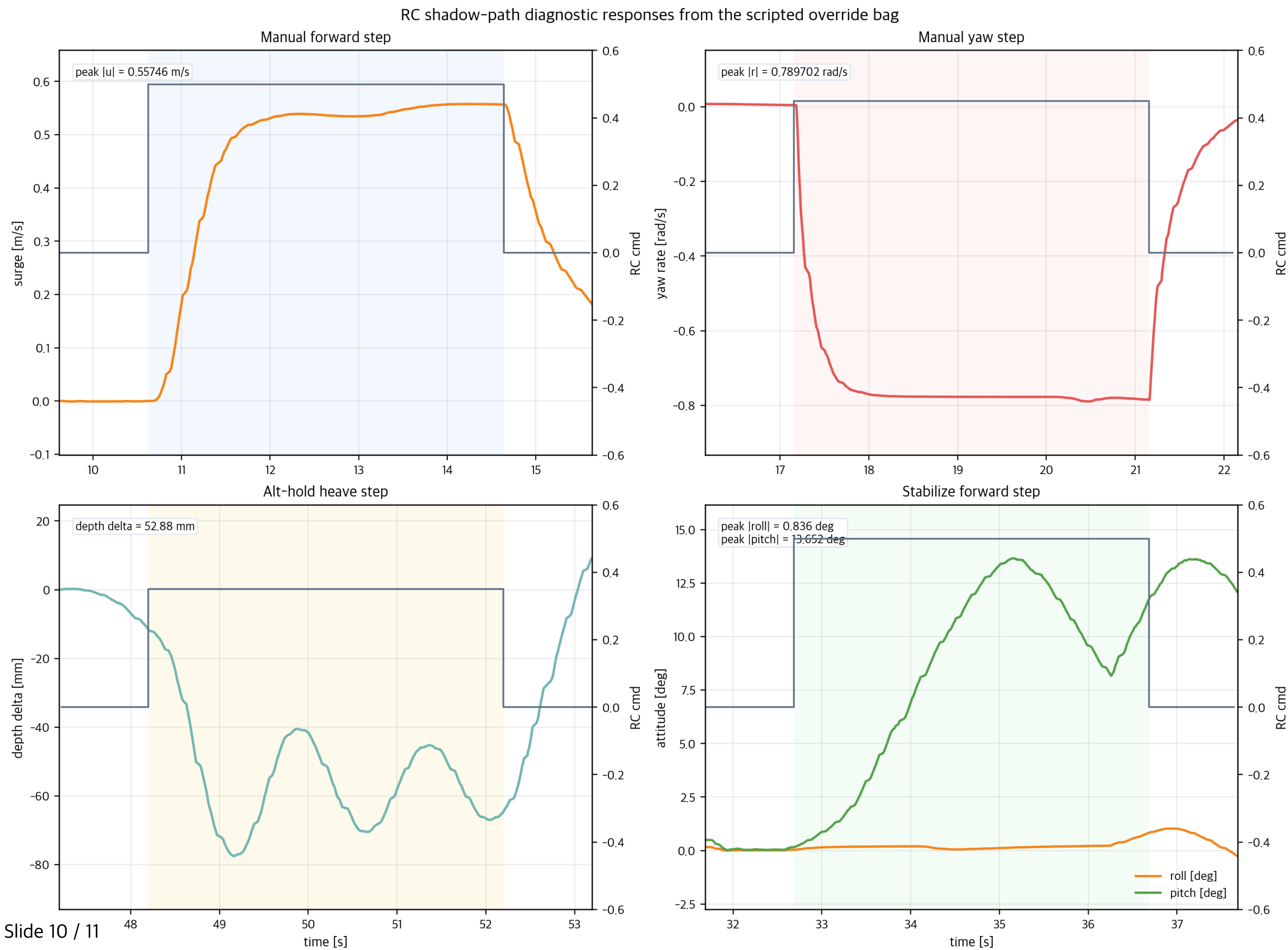
10. latest actual reference

- stabilize forward peak surge 0.557 m/s

- stabilize forward peak pitch 13.65 deg

- manual yaw peak rate 0.790 rad/s
- manual heave depth delta 42.34 mm

- current 분석의 실제 기준선은 이 latest rosbag



11. takeaways

- current = MuJoCo ellipsoid fluid + Python thrust/hydrostatics
- thrust는 simple polynomial model, buoyancy는 4-point distribution이 핵심
- 4-point buoyancy가 latest actual에 훨씬 가깝다
- 다음 튜닝 포인트: thrust authority, drag proxy, mode-dependent response

핵심 한 줄: current는 drag는 MuJoCo에, thrust와 hydrostatics는 Python에 맡긴 구조다.